

Technology Stack

Technology Stack Details :

Date	05 July 2026		
Team ID			
Project Name	A Comprehensive Measure of Well-Being		
Maximum Marks	2 Marks		
S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3	Used to create a responsive, userfriendly web interface for entering Human Development indicators and displaying the predicted HDI category.
2	Backend / Server-Side	Python, Flask	Flask provides a lightweight web framework for handling user requests, processing input data, and integrating the machine learning model efficiently.

3	Machine Learning Layer	Random Forest Classifier (Scikitlearn)	Used to train the Human Development Index (HDI) prediction model and classify countries into Very High, High, Medium, and Low Human Development categories
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4	Model Storage	Pickle (.pkl)	The trained Random Forest model is saved as hdi_model.pkl, enabling fast loading and prediction without retraining the model each time.
5	Development Environment	Anaconda, Jupyter Notebook, VS Code	Jupyter Notebook was used for data preprocessing, Exploratory Data Analysis (EDA), and model training. VS Code was used for developing the Flask web application.
6	<u>Libraries / Tools</u>	Pandas, NumPy, Matplotlib, Seaborn, Scikitlearn, Pickle	Pandas was used for data preprocessing, NumPy for numerical operations, Matplotlib and Seaborn for data visualization, Scikit-learn for model training and evaluation, and Pickle for saving/loading the trained model.